

Jonathan Kalsky

US Citizen | Junior Classification | Open to work from anywhere

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BIOGRAPHY

Computer Science student focused on software and systems engineering, with hands-on experience building automation across cloud platforms and real infrastructure. Experienced in developing reliable, observable, and performance-focused systems across backend, AI, and infrastructure environments.

Languages: C++, Python, Java, TypeScript/JavaScript, Shell, Swift, HTML, CSS, LaTeX, R, Haskell | English (Native), Hebrew (Native), Mandarin Chinese (~HSK2)

Experience: Software Engineering, Mobile App Development, Machine Learning, Web Development, Shell Scripting, Object Oriented Programming, Local LLMs, Agile Sprint Lifecycles, Scripting Arduino Embedded Systems, DSA, Memory/Space Complexity

Tools: Jira/Trello, React, VSCode, Xcode, IntelliJ IDEs, GitHub, Kaggle, Jupyter, Linux(processes, files, basic networking, scripting)

WORK EXPERIENCE

Platform Engineering Intern - Cisco Stack Automation - Quali [DevOps, Cloud, Automation]

[July 2026 - August 2026]

#Systems Automation

- Worked directly on building and improving infrastructure automation used to provision and manage environments across AWS, Cisco Intersight, servers, and hardware.
- Built preflight validation into workflows to catch invalid credentials, resources, and configuration before expensive provisioning operations.
- Diagnosed automation failures across infrastructure layers; one Terraform/vCenter teardown failure revealed a common underlying pattern across several automation blueprints, and this formed a basis for my solution that cut down costs and time to end users.
- Improved automation workflows and performance while validating changes against real infrastructure rather than simulated environments

LLM Hypervisor Research - Systems Researcher [Swift, SwiftUI, macOS]

[May 2026]

#Systems Research

- Designed and implemented a macOS **proof-of-concept** tool that **reallocates** CPU and memory **resources** from inactive applications to resource-intensive local LLM workloads
 - Investigated workload suspension, restoration, and orchestration to **optimize local inference performance**
 - Evaluated **resource management strategies** and system tradeoffs for dynamically scheduling local AI workloads on **restricted** hardware
- Cognitive Routines Research - Research Orchestration [Python, AntigraVity CLI, Agent Systems, GitHub Pages] [June 2026]
- #Agentic Research
- Conducted exploratory research investigating how system prompts **influence LLM behavior**, reproducibly evaluating behavioral specialization independent of model capabilities
 - Designed and evaluated a **cognitive routing methodology** that dynamically selects specialized reasoning workflows based on task characteristics and behavioral objectives
 - Published research artifacts, experimental setups, and findings through GitHub Pages to encourage peer review and investigation

NexoDeck - Education Platform - [React, TypeScript, Node.js, LaTeX]

[January 2026 - May 2026]

#Product Development

- Identified and **reduced** latency **bottlenecks** in an AI pipeline by **parallelizing** execution with promise **concurrency** and **context caching**
- Developed a **validated** LaTeX processing pipeline with checks that ensure consistent and **reliable** rendering
- Implemented logging and observability to trace API usage and system behavior, enabling performance **monitoring** and bottleneck detection

Web Applications -Full Stack Prototypes - [JavaScript, Python, React, AI-Accelerated]

[Hackathons 2025-2026]

#Quick Team Results

- Built and deployed full-stack applications under rapid development timelines, and debugged distributed application workflows
- Earned awards across multiple hackathons while collaborating in fast-paced teams and delivering production-ready demonstrations

Aggie Course Explorer - Full-Stack Course Planner - [React, JavaScript/TypeScript, Python, PostgreSQL]

[October 2025 - May 2026]

#Product Development

- Led development of React + TypeScript components for a production student course-planning platform serving Texas A&M students
- Designed parsing and visual graphing logic to represent object relationships, utilized across components; promoted to **team lead**
- Analyzed and **extended** existing **backend** data flows and REST APIs to support new features and improve system integration
- Developed CLI **tooling** to simulate backend workflows, accelerating **debugging** and **validation** of system interactions
- Improved **scalability** by replacing large JSON loads with a queryable database, reducing latency and memory overhead via efficient querying

Pixedify - iOS Photo Editor - [iOS App, AI Software]

[June 2025 - May 2026]

#Product Development

- Engineered asynchronous processing across OpenAI, Pinecone, and Firebase, coordinating image analysis, vector generation, and metadata persistence
- Implemented a Pinecone vector database to store user **embeddings** and improve **similarity search** results for recommendations

Apple Swift Student Challenge - Software Developer - [iOS App, Data Collection, ML]

[12th grade]

#Product Development

- Built an iOS application that trains and evaluates a handwritten digit **classification model** using collected training samples
- Collected and labeled training data via a custom UI to improve classification accuracy and **evaluate** model performance

TimeWize - Founder, Developer, Marketer, Publisher - [iOS App, Project Management]

[11th-12th grade]

#Teamwork

- Designed backend data pipelines using Firebase (authentication, real-time database, cloud storage)

Education

Texas A&M University

College Station, TX, 2025-2029

Bachelor of Science in Computer Science

Cedar Park High School

GPA: Weighted: 5.172 / Unweighted: 4.0

Top 10% | AP Scholar

Dual Enrollment

ACADEMIC AWARDS:

Dean's Honor Roll, Tidal Hack: 2nd Place beginner team, **Datathon:** 1st Place best use of ElevenLabs, 2nd Place in video-game-agent Reinforcement Learning, **Build4Good:** 1st Place Solo Hack, **Aggie Coding Club:** Best Project Member

Multi-District & Regional Qualifier (2023-2025): Earned four 1st-place team titles (CS, Math, Number Sense) and 2nd in Regional CS (2024)

Certifications in Java, Swift, R, and OSHA-10 Hour

Involvement: Aggie Coding Club, Hackathons, Computing Society, TamuHack, TIDAL (ML)

Current Growth Specialist (Part-Time), Quali — Cisco Stack Automation / Torque

TIMELINE

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